

Chapter 7

Masonry mechanical properties

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7.1 Introduction

Accurate modeling of the performance of masonry structures requires a thorough experimental description of its behavior and development of adequate constitutive laws that can represent its heterogenous behavior. The extent of knowledge required on materials properties depends on the modeling strategy and analysis method of interest. Elastic properties of the units and mortar or masonry as a composite is sufficient for linear analysis. However, masonry cracks at very low stress levels, which makes the reliability of linear analyses methods doubtful. For nonlinear analysis, in addition to the elastic properties, nonlinear mechanical properties should be established. Special attention should also be given to the overall aspects of the structure: unit dimensions, type and quality of mortar joint, and unit surface conditions (perforations or indentations). The details of the experimental method used for establishing the material properties, especially boundary conditions, are of critical importance and should be carefully considered in interpretation of the experimental results.

The large variety of units and mortar types and construction methods used worldwide do not allow development of unified constitutive laws. For this reason, several codes and standards prioritize experimental characterization of material properties for design and numerical simulations. However, in some cases either performing experimental tests is not possible at all (such as in historical structures) or a limited set of experimental tests are performed (usually compressive and flexural tests for characterization of strength) due to economical reasons or lack of suitable testing equipment.

This chapter discusses the way these mechanical properties are established and gives some suggestions for establishment of material properties for numerical simulations when limited experimental results are available.

Other mechanical properties like creep, moisture movement and thermal expansion, although extremely important, are beyond the scope of this chapter.

7.2 Masonry in general perspective

7.2.1 Type of units and materials

Masonry is made by piling units on top of each other in a certain pattern using mortar to allow for dimensional tolerance. In some cases, the units are connected with interlocking mechanisms without application of any mortar joint. The units are available in several dimensions and types (Fig. 7.1), with the smallest brick size being approximately $210 \times 100 \times 55 \text{ mm}^3$. These units can be laid using one hand. The next size are blocks, in sizes ranging from $200 \times 100 \times 100 \text{ mm}^3$ to $300 \times 600 \times 100 \text{ mm}^3$ and sometimes even larger depending on the volume mass of the material they are made of. The dimension is often limited by rules concerning the weight 14–15 kg max. Blocks are laid using both hands, in some cases by two persons. The largest are element-sized units of approximately $600 \times 900 \times 100\text{--}300 \text{ mm}^3$. This means only four layers are generally needed to build a wall of 2.5 m height. Besides dimensions, the other main distinction in masonry units is the materials they are made of. Fired clay and natural stone are the most common types all over the world. Other contemporary materials are calcium silicate, aerated concrete, and normal concrete. Units can be made quite easily by filling a prismatic mold with the paste. Small indentations (so called frogs) or perforations can also be made relatively easily. Some clay bricks and blocks are made with an extrusion process that allows for highly perforated units,



FIGURE 7.1 Some typical masonry units. *Photographs from producers brochuers.*

with perforations up to 50% of their section. Concrete blocks can be made with webs of ± 20 mm allowing for voids large enough to apply reinforcement with grouting.

Mortars are a mixtures of sand and a binder, historically lime. After cement became available it was often used in combination with lime resulting in the so-called bastard mortars. According to Eurocode 6 three types of mortars may be used: general-purpose mortar, thin-layer mortar, and light-weight mortar. The joint thickness depends on the dimensional tolerances and the unit size. The so-called general-purpose mortar allows a joint thickness between 10 and 15 mm for bricks and up to 20 mm for larger blocks. Moreover, calcium-silicate elements are made with smooth and level surfaces and with small dimensional tolerances that allow the use of thin mortar joints (2–3 mm thick).

The properties of a mortar before and after curing depend on its constituents like sand, cement, and/or lime and additives, and the amount of water used during building and available during curing. In general, lime-based mortars are more flexible (lower elastic modulus) and cement-based mortars are more brittle. Additionally, cement-based mortars harden faster than lime-based mortars.

It is clear that the type of masonry must be considered for numerical simulation. The modeling details and required parameters differ depending on the modeling strategy followed (see next section). The properties of units and mortars are the main parameters to be characterized. In modeling of the brick–mortar interaction, the surface properties (and perforations/indentations) of the units must also be considered. Depending on the production procedure, surfaces may be smooth or sanded. Sand is often used to allow for easy demolding. The surface structure is also clearly dependent on the unit type (e.g., surface characteristics of concrete are different than those of extruded clay). These characteristics are in principle considered in the constitutive law and properties of the interface. A detailed discussion on the properties of units and mortars can also be found in the literature (Hendry, 1990; Drysdale et al., 1993; Hendry et al., 2004; Klingner, 2010).

7.2.2 Modeling at different levels of detail

The set of mechanical properties required for simulation of the structural performance of masonry depends both on the range of interest (elastic or inelastic) of material's response and on the adopted modeling approach. For simplified modeling purposes (elastic range), knowing the modulus of elasticity (E) and the compressive strength (f_c) is usually sufficient. When linear elastic behavior is assumed, the secant E value for the part of the stress–strain diagram to one-third of the ultimate stress can be used. When masonry cracking occurs, nonlinearity effects become apparent. Modeling the cracking and failure processes in masonry structures requires additional

mechanical properties like shear strength, tensile strength, and fracture energies (G_f). The fracture energy is defined as the amount of energy required for creating a unit crack area, and it is obtained as the area under the stress–displacement diagram in the softening regime (postpeak). This definition is straightforward for tensile tests, but not for compressive tests. Aspects such as whether the compressive fracture is a material property or a structural property or the compressive fracture energy represents formation of several cracks and not a single crack are still under discussion. Depending on the constitutive law and the finite element (FE) package used, this value might be defined differently, but in most cases it is defined as the area under the stress–displacement diagram in the postpeak regime.

To obtain these parameters, detailed measurements in dedicated test set-ups must be performed. In these tests the main problem is applying the appropriate boundary conditions.

As for the modeling approaches, depending on the level of accuracy and the simplicity desired, the following three different strategies are usually used:

- Detailed micromodeling—units and mortar in the joints are represented by continuum elements, whereas the unit-mortar interface is represented by discontinuous elements. In this approach, Young's modulus, Poisson's ratio and, optionally, inelastic properties of both unit and mortar are required. The interface represents a potential crack/slip plane with initial dummy stiffness to avoid interpenetration of the continuum. The nonlinear properties of the interface are also required.
- Simplified micromodeling—expanded units are represented by continuum elements, whereas the behavior of the mortar joints and unit-mortar interface is lumped into discontinuous elements. In this approach, each joint, consisting of mortar and the two unit-mortar interfaces, is lumped into an average interface while the units are expanded in order to keep the geometry unchanged. Masonry is thus considered as a set of elastic blocks bonded by potential fracture/slip lines at the joints. Accuracy is lost in this modeling method since the Poisson's effect of the mortar is not included. Discrete element modeling (DEM), an approach that has recently received increasing attention, is also followed based on similar assumptions.
- Macromodeling—units, mortar, and the unit-mortar interface are smeared out in the continuum. This approach does not make a distinction between individual units and joints but treats masonry as a homogeneous continuum. Here, the elastic and inelastic properties of the masonry as a continuum are necessary. In this case, the damage of masonry is usually modeled following a smeared crack or damage plasticity approach.

7.3 Unit properties

The variability of the masonry units used around the world for construction of masonry and historical structures together with the different deterioration processes due to environmental action makes the proposal of unique relations for prediction of mechanical properties a complicated task. Old masonry units present large variation in their mechanical properties even if the samples are chosen from the same building. In contrast, new masonry units usually have a much lower variation due to the controlled construction procedures. A problem in evaluation of the mechanical properties of materials in existing and historical structures is that performing destructive tests is usually prohibited. This has led to the development of a range of nondestructive techniques for the evaluation of the mechanical properties. Estimation of the mechanical properties based on visual aspects is another technique proposed by the New Zealand Society for Earthquake Engineering (NZSEE) (Table 7.1).

Another important issue is that masonry units are usually assumed isotropic, although extruded units can present different mechanical properties along different directions.

The main elastic properties of bricks are the elastic modulus, E , and Poisson's ratio, ν . If the nonlinear response of the units is also considered the compressive strength, f_c , the peak strain, ε_u , the postpeak fracture energy in compression, G_f^c , the tensile strength, f_t , and the tensile fracture energy G_f^t are required. The compressive strength is probably the simplest parameter for experimental characterization and for this reason other mechanical properties are usually correlated with this value. The ratio between the fracture energy and the compressive strength is sometimes called the ductility index, d_{uc} , and can be used as an indication for the brittleness of the response (see Chiaia et al., 1998).

TABLE 7.1 NZSEE recommendations for bricks mechanical properties based on visual aspects.

Type	Description	f_c (MPa)	E (GPa)	ν
Soft	Weathered, pitted, distinct color variation with depth, bright orange, possibly under fired	1.0–5.0	4	0.35
Stiff	Common brick, can be scored with a knife, red	10.0–20.0	13	0.2
Hard	Dense, hard surface, well fired, dark reddish brown	20.0–30.0	18	0.2

7.3.1 Elastic properties

The elastic modulus can be obtained from measurements on pieces of units loaded in compression. The specimens have to be capped at their loaded ends to allow for a smooth force transfer for reducing peak stresses from the steel load platens into the specimen. Deformation sensors (e.g., LVDTs) are mounted on the specimen to measure the deformation. The deformation should be measured in an area sufficiently far away from the loaded ends due to the three axial stress states in these areas. The specimens may be assumed to be in a uniaxial condition far from the loaded ends. Fig. 7.2 shows LVDTs with a measuring length of 185 mm (three mortar joints, unit height = 52 mm and joint height = 13 mm) on a calcium silicate masonry prism. Stresses are obtained by dividing the measured forces by the loaded gross sectional area and the strains by dividing the measured displacement of the end points of the measuring device by their distance. Usually the samples are loaded in a range of stresses far from the compressive strength (to avoid damage in the specimens) for several cycles. The average slope of the loading regimes is then accounted as the elastic modulus. The specimens are either made while building the real masonry or afterward from the same constituents.

At the end of compression testing, the measuring devices may fall off the specimen, which makes it complicated to record the postpeak behavior. Then the results of the load platen movement may be used. A correction factor, however, may be needed in these cases. Fig. 7.2 shows an almost ideal situation where LVDTs measured reliably until the end of the test.

When experimental results are not available, the elastic modulus can be estimated from the compressive strength of the unit. However, it should be noted that a wide range of relations can be found in the literature for this purpose showing that units properties vary significantly with type and

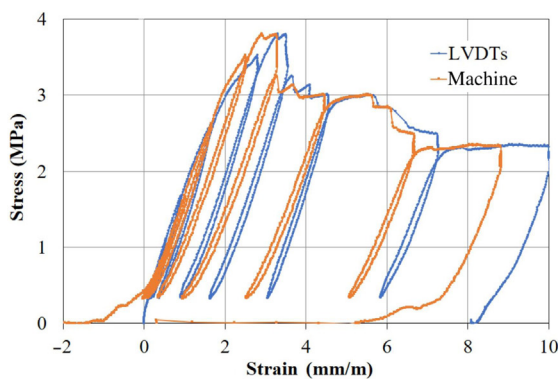


FIGURE 7.2 Example of a stress–strain diagram from a compression test on a 70×70 cm Wallette with cyclic loading showing the difference between measuring load platen displacement and deformation of the central part of the specimen.

production method. The NZSEE proposes (Table 7.1) a range of 125–1400 f_c with an average value of 350 f_c for historical bricks. This average is very similar to the relation proposed for North Indian bricks in Kaushik et al. (2007). Caporale et al. (2015) reported an elastic modulus of 160 f_c for adobe bricks and Vasconcelos (2005) found a value of 380 f_c for granite stone. In the Dutch study (CUR 171) a ratio of 420 f_c is given for solid clay, soft mud, and extrusion bricks, with E values of 6 GPa for soft mud units ranging to 16.7 GPa for the strongest extrusion units. These values were obtained by measurements on piles of five units in height bonded with a two-component-high strength material with thin joints (Vermeltoft, 2005).

7.3.2 Compressive behavior

Compressive strength tests are easy to perform. A large database of experimental compressive test results can be found in the literature. However, most of these studies are focused on derivation of the compressive strength and little information is available on the postpeak behavior necessary for extraction of the nonlinear parameters. Although it can also be argued that compressive testing is not suited to establish the compressive fracture energy as even under compression, the failure is due to exceeding the local tensile stresses.

Among the available experimental results, different test conditions including specimens' dimensions and moisture content, boundary conditions, and temperature have been adopted which can significantly affect the reported results. These differences make comparisons between different experimental results complicated. If the test conditions are known, it is suggested to convert the results to an equivalent compressive strength relevant to the air-dry conditioning regime. For example, for calcium silicate, this value can be obtained by multiplying the compressive strength by 0.8 if the units are oven dried and by 1.2 if they are submerged in water for over 15 hours. The compressive strength of well-fired clay bricks is generally not sensitive to moisture content, but nonwell-fired bricks and stone are sensitive to moisture content. The boundary conditions may also affect the results. The friction between the load platen and the specimen produces a triaxial stress state near the boundaries and prevents free expansion of the specimen. This, depending on the specimens' shape and size, can affect the observed strength and the postpeak response (thus affecting the fracture energy). The use of two sheets of Teflon reduces this friction considerably. For comparison of results of specimens of various sizes, shape factors can be used. A shape factor gives the ratio between a compression test result on a specimen of a certain dimension and a reference specimen presumably made of the same material. Several equations exist, from which the equation of Khalaf and Hendry (1989) is quite convenient for evaluating this correction factor:

$$C100 = (h/A^{0.5})^{0.37}$$

where C100 is the correction factor with a cube with ribs of 100 mm as reference, h is the height of the specimen, and A is the loaded area.

The common range of compressive and tensile strength values for different types of units are presented in [Tables 7.2 and 7.3](#). It can be observed that the compressive strength of the units varies significantly with the type. In general, ancient masonry units present lower mechanical properties with larger variability compared to new units. It should be noted that in fired clay bricks the compressive strength is highly dependent on the firing temperature.

If the experimental data on the postpeak compressive response is not available, the fracture energy can be estimated from the compressive strength

TABLE 7.2 NZSEE reference values for compressive and tensile strength of masonry units and mortar.		
Material	Compressive strength (MPa)	Tensile strength (MPa)
Natural stone	10–200	1–5
Clay brick	10–70	1
Calcium silicate	10–35	1–2
Concrete	10–35	3
Aerated concrete	3–7.5	0.5–2
Gypsum	5–10	1–2
Loam (soft mud)	2–3	0–0.33
Mortar	3–25	$0.25–0.33 \times f_c$

TABLE 7.3 Typical ancient masonry unit types.	
Materials	f_c (MPa)
<i>Stone</i>	
Igneous—granite	40–150
Sedimentary—limestone	10–100
Metamorphic—marble	30–150
Metamorphic—schist	5–60
<i>Clay</i>	1.5–30
<i>Adobe</i>	0.5–3

of the unit. Due to the lack of sufficient literature data on the nonlinear behavior of masonry units, comprehensive data on the correlation of the fracture energy with the compressive strength are not available yet. However, the formulation proposed for concrete in the Model Code 90 may be used. This code proposes a peak strain of 0.2% and a compressive fracture energy of $G_f = 15 + 0.43f_c - 0.0036f_c^2$. It should be noted that this formula is only applicable for compressive strength values between 12 and 80 MPa. For f_c values lower than 12 MPa, a ductility index, $d_{u,c}$, equal to 1.6 mm (MC 90 value for $f_c = 12$ MPa) and for f_c values higher than 80 MPa, a $d_{u,c}$ value equal to 0.33 mm is suggested (MC 90 value for $f_c = 80$ MPa). The fracture energy in granite stones can be obtained as $G_f = 2.54f_c^{0.58}$ as suggested in Vasconcelos (2005).

7.3.3 Tensile behavior

Tensile response of units can be obtained by performing direct tensile tests. Due to the complexities related to preparation of dog-bone shape specimens with these materials, the tests are usually performed on notched prisms. These tests are complex and the results, especially in the postpeak range, depend on the boundary conditions. At low load levels, it is assumed that stresses are uniformly distributed and the line of action of the load coincides with the center of gravity of the loaded area. At higher levels, stresses become larger than the strength of the material and (micro) cracking starts. Consequently, stress distribution will no longer be uniform and bending effects increase, locally leading to larger stresses and a brittle failure. These issues with direct tensile testing has lead to development of test setups in which the rotation of the specimen's ends can be prevented, for example, by Hordijk (for concrete) and Van der Pluijm (for masonry). As direct tensile tests are tricky to perform and usually large variations are obtained in the results, tensile splitting or flexural tests that require less preparation are preferred. However, it should be noted that the tensile strength obtained from these tests is, in principal, different than that of obtained from direct tensile tests.

Extensive information exists in the literature regarding the tensile strength and less on tensile fracture energy of masonry units (Van der Pluijm, 1999; Lourenço et al., 2005; Vasconcelos, 2005). Again, unified predictive relations are complicated due to the large variety of the unit types, manufacturing processes, specimens' dimensions, and testing conditions and procedures. Detailed studies show a dependency of tensile strength on production methods. In an extrusion process, the soft clay is squeezed through an opening, which brings the flat-shaped clay particles into layers, leading to the units having different properties in different directions (Vermeltfoort, 2005). The way clay particles are oriented in a brick, both soft mud—randomly—and extruded—more ordered—influences the strength, stiffness, and

failure pattern of a specimen. This is why the base material of extruded perforated units has almost linear behavior until fracture and is relatively strong, but brittle.

As a correlation with the compressive strength, it has been found that the brick tensile strength is in the range of 12%–14% of its compressive strength. For unreinforced adobe bricks a value of 36% of the compressive strength was proposed (Caporale et al., 2015). NZSEE proposes a tensile strength equal to 10% of the compressive strength for historical bricks. In The Netherlands a ratio of 18 between compressive (5.5%) and tensile splitting strength is often used for soft mud units and a ratio of 14 (7%) for extruded bricks.

The ratio between the tensile fracture energy, G_f , and tensile strength, f_t (defined as the ductility index, $d_{u,t}$), is around 0.018–0.040 mm for bricks (Van Der Pluijm, 1997, 1999; Lourenço et al., 2005). In the absence of specific information regarding the properties of the unit, using the average value, 0.029, is recommended. For stone granites, the ductility index is proposed as a function of tensile strength as $0.239 f_t^{-1.138}$ as suggested in Vasconcelos (2005). Assuming a tensile strength of 3.5 MPa, this relation gives a value of 0.057 mm for the ductility index, being the double of the value reported for the bricks. The Model Code 90 recommends $G_f = 0.025 (f_c/10)^{0.7}$ for the tensile fracture energy of concrete with a maximum aggregate size of 8 mm. Assuming that the tensile strength of masonry units is around 5% of the compressive strength on average, this expression can be rewritten as $G_f = 0.04 f_t^{0.7}$. This gives a fracture energy of 0.0976 N/mm and a ductility index of 0.028 mm for brick with an average tensile strength of 3.5 MPa.

7.4 Mortar properties

The most common binders used in masonry structures are cement-based or lime-based or a mix of the two. Lime-based mortars are either aerial lime or hydraulic lime. The aerial lime needs air for hydration and hardening, while hydraulic limes harden with water. Cement-based mortars can be found in both new and historical masonry structures, but lime-based mortars are usually found in historical masonry or in repair mortars used for historical structures.

The mechanical properties of mortars are dependent on several factors including the type and proportion of the mortar constituents as well as the curing and the testing (e.g., specimen geometry and test setup) conditions. The moisture transfer between the mortar and brick and the level of degradation can also significantly affect the mechanical properties of mortars. Often the quality (and therefore the mechanical properties) of mortar varies even within the same structure.

Experimental testing is therefore highly recommended for characterization of mortar mechanical properties. However, it should be noted that the experimental tests are usually performed on specimens with geometrical details and curing conditions that are different from those of mortar joints in real structures. Differences in the obtained mechanical properties should therefore be expected. If evaluation of material properties in an existing structure is of concern, often taking samples is not possible at all or the obtained samples are highly irregular and therefore tests need to be performed with consideration of these irregularities.

Mortars are usually classified in different categories depending on their expected mechanical properties. For example, ASTM 270 classifies mortars into four groups (M, S, N, and O mortars); see Table 7.4. The NZSEE provides benchmark mechanical properties for historical mortars; see Table 7.5. In The Netherlands, the graph shown in Fig. 7.3 is often used to explain the behavioral differences between the three main mortar types. It can be clearly observed that the lime-based mortar has a very slow hydration rate and lower compressive strength compared to that of cement and bastard mortars.

Despite the existence of a large database of experimental results in the literature, the above-mentioned issues make proposal of uniform relations or constitutive laws for predicting the in situ mechanical properties of mortars complicated. Additionally, the experimental specimens are generally made in laboratory molds and thus the results are not representative of real structural conditions. The experimental results are also often reported without detailed explanations on influencing factors including the test methods and curing conditions. Hence only a few studies can be found with sufficient details and direct use of the experimentally measured values in numerical simulations.

TABLE 7.4 ASTM C270 mortar types.

Type	Volume proportions			f_c (MPa)
	Portland cement	Hydrated lime	Sand	
M	1	Min: 0, max: 0.25	Not less than 2.25 nor more than three times some of the total volume of cement and lime	17.2
S	1	Min: 0.25, max: 0.5		12.4
N	1	Min: 0.5, max: 1.25		5.2
O	1	Min: 1.25, max: 2.		2.4

TABLE 7.5 Recommended mortar properties in NZSEE.

Type	Description	c (MPa)	μ	f_c (MPa)	E (GPa)	ν
Non cohesive	Lime-based, heavily leached and weathered, sand like, easily raked out by hand, unbound aggregate	0.0	0.0	0.0	0.0	0.0
Soft	Lime binding mildly leached, can be raked out, punch test <30 mm	0.1	0.4	1.0	7.0	0.05
Firm	Lime-based, not weathered, punch test <20 mm	0.2	0.6	4.0	9.0	0.07
Stiff	High Portland cement content, punch test <10 mm	0.4	0.8	8.0	12.0	0.11

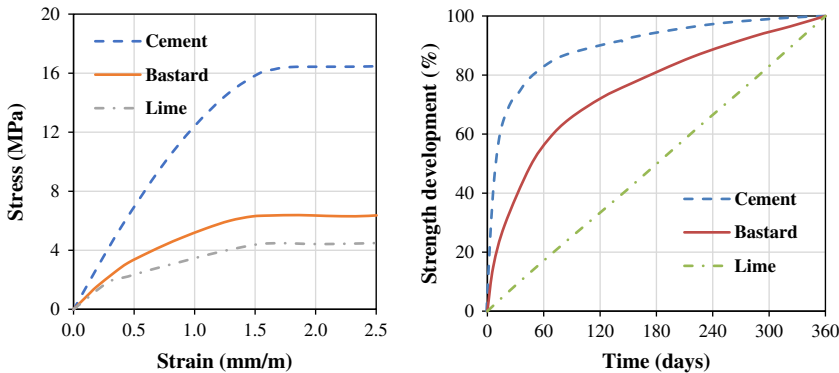


FIGURE 7.3 Characteristics of cement mortar, cement-lime mortar, and lime mortar.

Similar to the units, the main elastic mechanical properties of mortars required for numerical simulations are elastic modulus, E , and Poisson's ratio, ν . As for the nonlinear properties, the compressive strength, f_c , peak strain, ε_u , postpeak fracture energy in compression, G_f^c , the tensile strength, f_t , and the tensile fracture energy G_f^t are required.

7.4.1 Elastic properties

Testing of elastic properties of mortars can be done following the same procedure explained for the units. The available studies in the literature usually

lack characterization of elastic modulus of mortars. If performing experimental tests is not possible, the use of common formulations for concrete is recommended.

7.4.2 Compressive behavior

As for the masonry units, most of the available literature reports on the compressive strength of mortars and little information is available on the postpeak compressive behavior. Compressive tests are easy, but usually performed on half prisms left from three-point bending tests with the main focus on characterization of mechanical strength. Surface hardness measurements have also been extensively reported for in situ mechanical characterization of mortars. In cases where experimental results are not available, the proposal of Model Code 90 for concrete with maximum aggregate size of 8 mm can be used ($G_f = 15 + 0.43f_c - 0.0036f_c^2$).

7.4.3 Tensile behavior

Tensile tests are also mainly limited to characterization of the tensile strength. Similar to units, usually flexural tests or tensile splitting tests are preferred due to the complexities involved in performing direct tensile tests. Again these values should be converted to direct tensile strength with appropriate factors. Measurement of the postpeak response is also usually neglected due to the brittle behavior of the mortar. In such cases, the formulation proposed by Model Code 90 ($G_f = 0.025(f_c/10)^{0.7}$) can be used.

7.4.4 Leveling mortar

In experiments, sample specimens and walls are usually prepared and built outside the test setup. For this reason, a filling material is usually used to allow for a smooth load transfer during the tests and to make sure that the contact between the (steel) load platen and the masonry is snug and tight. In this way, the dimensional tolerances and the usually rough and grooved specimen's surface are bridged. In some situations, soft board serves this purpose quite well. A stiffer contact is obtained by using mortar. However, mortar takes some time to cure. Thus, gypsum could be an alternative.

This mortar, depending on its properties and the measurement details, might affect the obtained results. When simulating the response of experimental tests, the presence of leveling mortar should, therefore, be considered. The properties of this mortar can be established in a similar way as done for mortars to build specimens.

7.5 Mortar—brick interface properties

The interface bond between units and mortar is usually the weakest link in masonry structures and controls the failure mode in many cases. Its behavior is highly dependent on the brick properties (composition, strength, size,

surface with or without sanding, etc.); mortar properties (composition, water content, curing conditions, etc.) and workmanship (condition of joints, etc.); and the distribution of mortar over the contact surface and the consequent edge effects (Vermeltfoort, 2005).

The interface properties are usually required in micromodeling or DEM approaches where the interface elements at the unit-to-mortar or unit-to-unit interfaces. The input parameters are mainly the stiffness and nonlinear properties of the interface in the normal (mode I) and the shear (mode II) directions. These properties might be given as discrete values that are used internally by the FE package for generation of the bond-slip behavior. It is also possible in most cases to directly provide the interface laws.

The main nonlinear properties in the normal (tensile) direction are the bond tensile strength, f_t , and the bond fracture energy, G_f . Meanwhile, the bond strength, c (also called cohesion), the friction angle, $\tan \varphi$, and the dilatancy angle, $\tan \psi$, are generally needed in the shear direction. More advanced material models may require variable friction and dilatancy angles yielding to a nonconstant mode II fracture energy, however, these parameters can be assumed constant for general applications.

7.5.1 Elastic properties

The stiffness of interface elements in the shear, K_s , and normal, K_n , directions can be obtained from the elastic modulus (E_u and E_m) and shear modulus (G_u and G_m) of unit and mortar as $K_s = G_u G_m / t_m (G_u - G_m)$ and $K_n = E_u E_m / t_m (E_u - E_m)$. Here t_m is the mortar thickness. It should be noted that the interface is often a weak/soft spot. The above formulation might lead to negative or extremely low E -values in some cases (CUR 173) when subtracting the calculated brick deformation from the measured masonry deformation obtained from the experimental tests.

7.5.2 Tensile behavior

Direct tensile tests can be performed to obtain the bond properties under tensile loading (mode I) as shown in Fig. 7.4. Deformation controlled tests need to be carried out on masonry specimens to obtain the tension softening curve for calculation of the fracture energy. This fracture energy is defined as the amount of energy needed to create a unit area of a crack along the brick–mortar interface. It is equal to the area under the tension softening curve for a discrete crack. The values result from an extrapolation of the measured net bond surface of the specimen to the assumed net bond surface of the wall, neglecting any influence of the vertical joints. It should, however, be noted that the tensile bond response is highly brittle and measurement of the postpeak response is a complicated task even if the tests are run under displacement-controlled conditions.

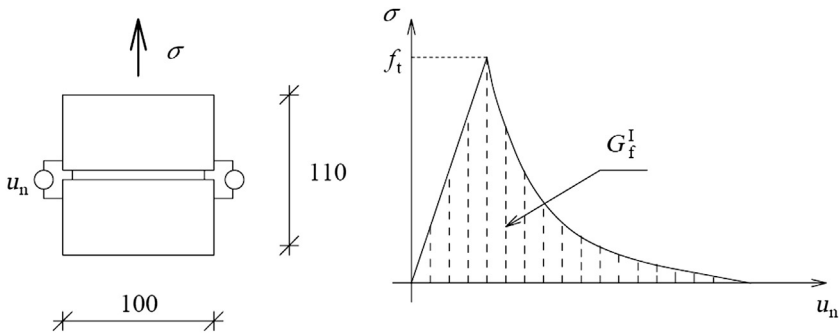


FIGURE 7.4 Test setup for mode I characterization of brick–mortar interface (Lourenço, 1994).

No specific recommendation can be given for obtaining these parameters based on the unit and the mortar properties. As a rough estimation, EC6 states that the tensile bond strength is generally in the range of 0.1–0.4 MPa. In Rots et al. (1997), the bond fracture energy was obtained in the range of 0.006–0.08 Nmm/mm² for a range of mortar and unit combinations. This value on average was about $c/10$ (0.012 Nmm/mm²) and can be used as a reference when experimental results are not available. Note that c is the bond strength (cohesion) and is explained in the next section.

7.5.3 Shear behavior

Triplet tests can be used to obtain the bond properties under shear loads (Fig. 7.5). As the shear response is dependent on the normal stress, the specimens should be tested at least under three different pre-compression levels. The shear and accompanying precompression loads are then plotted in a graph. The initial shear strength, that is, the shear strength at zero prestress can then be found by a linear regression of the experimental results to obtain the Mohr-Coulomb relationship. For repetitive loading, the residual stresses can be used to establish the coefficient of friction. When the specimen is loaded repetitively, a stick and slip behavior may occur when the direction of movement changes.

It should be noted that a triplet shear test is actually a four-point bending configuration and the stresses are not uniformly distributed over the height of the specimens during the tests. However, the relatively small span-to-height ratio of the specimens limits the bending moments. In such conditions, the simplifying assumption of uniform stress distribution can be reasonable. Alternatively, the shear test setup proposed by Van der Pluijm (1999) that was developed for minimization of the bending component can be used.

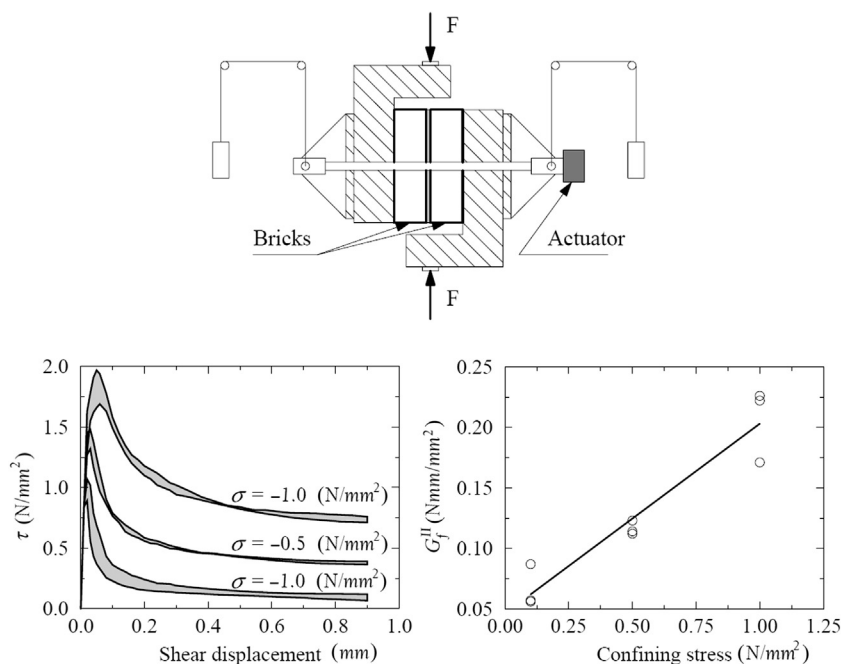


FIGURE 7.5 Load time diagram of shear test. Triplet shear test set up (Lourenço, 1996).

Van der Pluijm (1999) found that the ductility index, $d_{u,s}$, defined as the ratio between the fracture energy G_{II} and the bond strength c is in the range of 0.062–0.147 mm for different combinations of units and mortars. The recommended ductility index in the absence of more information is the average of 0.093 mm. It is noted that the Mode II fracture energy is clearly dependent on the normal stress level and the above values hold for a zero normal stress. The bond shear strength, c , is recommended to be in the range of 0.1–0.4 MPa in EC6. The formula proposed by Lumantarna (2012) that correlates the bond shear strength with compressive strength ($c = 0.055 f_c$) for historical masonry assemblages in New Zealand might also be used as an estimation.

The tangent of friction angle, $\tan \varphi$, can be assumed equal to 0.75 independent of the type of unit and mortar (Atkinson et al., 1989; Rots et al., 1997). The recommendation of EC6 is 0.4 and Augenti and Parisi (2011) proposed a value of 0.29 for the interface between tuff stones and hydraulic lime mortar. In some cases, the residual friction angle is also necessary and can be assumed equal to the friction angle. The dilatancy angle can also be assumed equal to zero as it diminishes with increasing axial stresses and slip at the interface.

7.6 Masonry properties

The main mechanical properties of masonry as a composite material are elastic modulus, Poisson's ratio, compressive strength, tensile strength, shear

strength, and corresponding fracture energies. Depending on the damage models and constitutive laws adapted, additional parameters might be needed. These parameters should be usually identified and selected based on the theoretical backgrounds of the selected constitutive law.

7.6.1 Elastic properties

The procedure for characterizing the elastic modulus of masonry is similar to that explained for units. The measuring area in this case should cover two or more bed joints and have a minimum length equal to $n \times$ layer height, with $n \geq 2$ where layer height is equal to the height of a unit plus the height of a joint.

Again, an extensive effort has been devoted to propose formulations for correlating this parameter to the compressive strength of masonry prisms. These formulations might be used when experimental results are not available. However, this should be done with special care as the range of proposed factors is very wide. EC6 proposes to obtain the elastic modulus as the secant modulus from testing or directly from the compressive strength as $1000f_k$. The National Annex to EC6 by NL proposes a value of $700f_k$. Kaushik et al. (2007) obtained a range of $250\text{--}1100f_k$ with an average of $550f_k$ for masonry prisms made of four different North Indian bricks and three mortar grades. Similarly, Drysdale et al. (1993) reported a wide range of $210\text{--}1670f_k$. As the result of compressive tests on masonry prisms made of historic and new clay bricks obtained from existing buildings, Vermeltfoort and Martens (2016) obtained a ratio of the elastic modulus of $240f_k$.

7.6.2 Compressive behavior

The compressive stress–strain diagrams are one of the main inputs for numerical simulations. Linear, parabolic-rectangular, rectangular, idealised or parabolic stress–strain diagrams have been frequently used (Fig. 7.6). For masonry made of (highly) perforated units, it is recommended to use a linear elastic no-tension diagram due to the brittle behavior of such masonry. Although a bilinear diagram is not usually suggested, using this diagram makes calculations much easier compared to the adopted parabolic-rectangular diagrams. Although, with the recent advancements in the available FE programs implementation of all these constitutive laws can be done with little effort.

The shape of the stress–strain curve or its main characteristics (compressive strength and fracture energy) can be directly obtained from experiments. These experiments can be easily done for new masonry where the unit and mortars are available.

Historic masonry (or existing buildings) can be examined by cutting pieces of masonry from existing buildings or by using in situ test procedures.

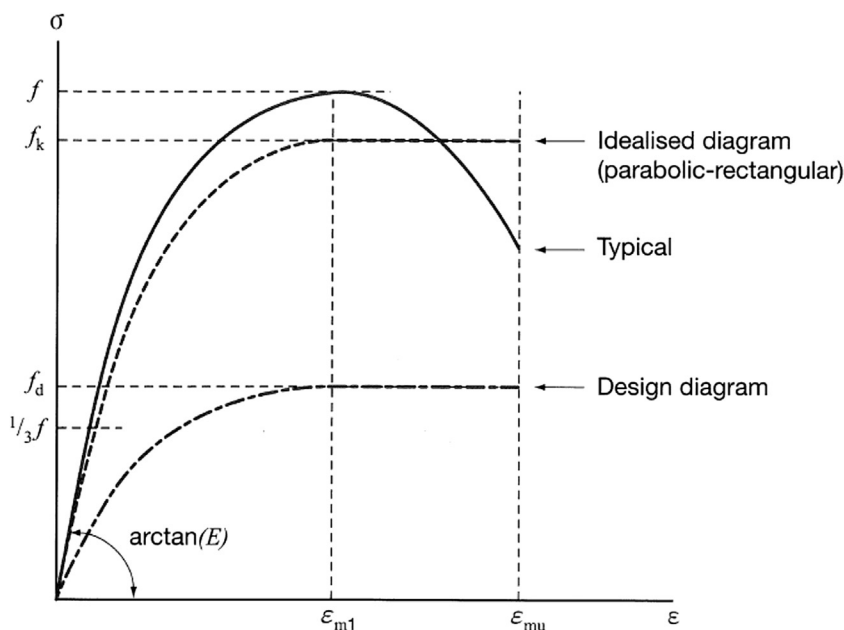


FIGURE 7.6 Parabolic rectangular stress–strain diagram for masonry.

For example, the tests performed by Vermeltfoort and Martens (2016) on pieces of masonry cut from existing walls (Fig. 7.7). When only units are available, which is often the case for historic lime mortars with relatively small strength, specimens can be made using these old bricks and a replica of the original mortar.

Several analytical formulas have been proposed for prediction of the masonry compressive strength based on the properties of units and mortar. These formulas are generally in the form of $f_k = K f_b^\alpha f_m^\beta$ where f_b is the units' compressive strength and f_m is the mortar compressive strength. EC6, for example, proposes this formula with different sets of parameters for different combinations of masonry and mortars. As an example, these are proposed as parameters $K = 0.6$, $\alpha = 0.65$, and $\beta = 0.25$ for solid units and general-purpose mortars.

The values proposed by PIET-70 for compressive strength of stone masonry depends on the stone type and are in the range of 0.5–8 N/mm² (see Table 7.6). Sarhat and Sherwood (2014) modified this formula as $f_k = 0.886 f_b^{0.75} f_m^{0.18}$ for hollow concrete block masonry systems made of concrete blocks with compressive strength in the range of 9–50 MPa. It should be noted that these relations are only valid for compressive strength of masonry prisms in the perpendicular direction to the horizontal joints. As masonry is an orthotropic (or anisotropic) material, it has different mechanical properties under different directions. The available experimental results



FIGURE 7.7 Stack-bonded prisms for detailed observation of behavior under compression and (right) wall piece after testing.

on the compressive strength of masonry under different directions is still limited. In the case of lack of experimental data, the compressive strength of solid unit masonry in the direction of horizontal joints can be assumed as $0.75f_k$ (Glitzka, 1988). For the compressive fracture energy, the values proposed for concrete in the Model Code 90 (CEB-FIP, 1993) may again be used ($G_f = 15 + 0.43f_c - 0.0036f_c^2$). EC6 proposes a ratio of fracture energy to elastic modulus of $G_f/E = 0.4$ for clay and lightweight concrete masonry systems that can be used when experimental results on the elastic modulus are available.

7.6.3 Tensile behavior

For tensile loading, theoretically three failure mechanisms can be observed in masonry prisms perpendicular to the bed joints: (1) tensile failure of the brick or mortar; (2) tensile failure at the mortar–brick interface; and (3) tensile failure at the brick–mortar interface in horizontal and vertical joints. When the first failure mechanism occurs, the tensile strength of the masonry can be assumed equal to the brick’s or mortar’s tensile strength considering the material that has failed. The third failure, failure at the brick–mortar interface, is the most common failure mode. In this case, the masonry tensile strength can be assumed to be equal to the tensile bond strength (Klingner, 2010). Some experimental results show that this value is in the range of 10%–20% of the masonry compressive strength (Hamid and Heidebrecht, 1979; Klingner, 2010). This is actually a very rough

TABLE 7.6 Design compressive strengths recommended by PIET-70 for stone masonry.

Stone type	Stone strength (N/mm ²)	Ashlar masonry			Other masonry		
		Dry, with good adjustment between faces	Ashlars h > 0.30 m Mortar > M8	Ashlars h < 0.30 m Mortar > M4	With well-defined courses Mortar > M4	Irregular Mortar > 0.5	Dry
Granite Basalt	>100	8.0	6.0	4.0	2.5	1.0	0.7
Quartz sandstone Hard limestone Marble	>30	4.0	3.0	2.0	1.2	0.8	0.6
Lime sandstone Soft limestone	>10	2.0	1.5	1.0	0.8	0.6	0.5

estimation and may not be accurate in all cases. As for the fracture energy, the formulation proposed for the units and mortar may be adapted in the lack of experimental results.

7.6.4 Shear behavior

Shear is a complex phenomenon and several methods of testing are available for its characterization. The masonry shear strength can be obtained as ($f_{vk} = f_{vk0} + 0.4\sigma_d$, $f_{vk} < 0.065f_b$, $f_{vk} < f_{vlt}$), where f_{vk0} is the masonry cohesion or initial shear strength (takes a value of 0.2–0.6 MPa depending on the joint type); 0.4 is the tangent of frictional angle (for safety reasons, a relatively small value is taken for this parameter); σ_d is the compressive stress on the masonry; and f_{vlt} is a limit value that can be taken from EC6. The masonry cohesion, f_{vk0} , is generally in the range of 0.15–0.3 MPa for new masonry and 0.05–0.1 MPa for old masonry with weak mortars. The tangent of frictional angle is recommended to be taken as 0.3 (for irregularly coursed stone) and 0.2 (for rubble masonry) in old masonry structures (Vasconcelos, 2005).

7.7 Overview of the input parameters for numerical simulations

As an overview, the main mechanical properties required as the input of numerical simulations and suggestions for when experimental results are not available are presented in the following tables (Tables 7.7–7.9).

TABLE 7.7 Unit and mortar main mechanical properties.

Parameter	Description	Recommendation
E	Elastic modulus	350 f_c for historical bricks
		380 f_c for stone
		160 f_c for adobe bricks
f_c	Compressive strength	Tables 7.1–7.3
f_t	Tensile strength	7%–10% f_c for bricks
		5% f_c for soft mud
G_i^c	Fracture energy in compression	$15 + 0.43f_c - 0.0036f_c^2$
G_i^t	Fracture energy in tension	$0.025(f_c/10)^{0.7}$ or $0.029 f_t$

TABLE 7.8 Interface properties.

Parameter	Description	Recommendation
K_s	Normal stiffness	$G_u G_m / t_m (G_u - G_m)$
K_n	Shear stiffness	$E_u E_m / t_m (E_u - E_m)$
c	Bond strength (joint cohesion)	$0.055 f_c$, 0.1–0.4 MPa
$\tan \varphi$	Tangent of friction angle	0.4–0.75
$\tan \varphi_r$	Tangent of residual friction angle	$\tan \varphi$
$\tan \psi$	Joint dilatancy	0
f_t	Joint tensile strength	0.1–0.2 MPa
G_f^I	Mode I fracture energy	0.1 c Nmm/mm ²
G_f^{II}	Mode II fracture energy	0.1 c Nmm/mm ²

TABLE 7.9 Masonry mechanical properties.

Parameter	Description	Recommendation
E	Elastic modulus	300–700 f_c
f_c	Compressive strength	$0.6 f_b^{0.7} f_m^{0.3}$
f_t	Tensile strength	10%–20% f_c
f_v	Shear strength	$f_{vk} = f_{vk0} + 0.4 \sigma_d$
G_f^c	Fracture energy in compression	$15 + 0.43 f_c - 0.0036 f_c^2$
G_f^t	Fracture energy in tension	$0.025(f_c/10)^{0.7}$ or $0.029 f_t$

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